11. $A=\begin{bmatrix}1&2&4\\0&1&5\\-2&-4&-3\end{bmatrix}$, $b=\begin{bmatrix}-2\\2\\9\end{bmatrix}$

By Theorem 3, the equation $Ax=b$ has the same solution as the system of equations corresponding to the augmented matrix $\begin{bmatrix}A & b\end{bmatrix}$.

$$\begin{bmatrix}1&2&4&-2\\0&1&5&2\\-2&-4&-3&9\end{bmatrix}\xrightarrow[+2\times Row1]{Row3=Row3}\begin{bmatrix}1&2&4&-2\\0&1&5&2\\0&0&5&5\end{bmatrix}\xrightarrow[Row2-Row3]{Row2=}\begin{bmatrix}1&2&4&-2\\0&1&0&-3\\0&0&5&5\end{bmatrix}$$

$$\xrightarrow[-Row3]{Row1=Row1}\begin{bmatrix}1&2&-1&-7\\0&1&0&-3\\0&0&5&5\end{bmatrix}\xrightarrow[+\frac{1}{5}\times Row3]{Row1=Row1}\begin{bmatrix}1&2&0&-6\\0&1&0&-3\\0&0&5&5\end{bmatrix}\xrightarrow[\times\frac{1}{5}]{Row3=Row3}$$

$$\begin{bmatrix}1&2&0&-6\\0&1&0&-3\\0&0&1&1\end{bmatrix}\xrightarrow[-2\times Row2]{Row1=Row1}\begin{bmatrix}1&0&0&0\\0&1&0&-3\\0&0&1&1\end{bmatrix}$$

The corresponding eqns are: $x_1=0$, $x_2=-3$ and $x_3=1$.

$$Ax=b \Rightarrow \begin{bmatrix}1&2&4\\0&1&5\\-2&-4&-3\end{bmatrix}\begin{bmatrix}0\\-3\\1\end{bmatrix}=\begin{bmatrix}-2\\2\\9\end{bmatrix}$$

By defn. of Ax, $0\begin{bmatrix}1\\0\\-2\end{bmatrix}+(-3)\begin{bmatrix}2\\1\\-4\end{bmatrix}+1\begin{bmatrix}4\\5\\-3\end{bmatrix}=\begin{bmatrix}-2\\2\\9\end{bmatrix}$

12. $A=\begin{bmatrix}1&2&1\\-3&-1&2\\0&5&3\end{bmatrix}$, $b=\begin{bmatrix}0\\1\\-1\end{bmatrix}$

By Theorem 3, the equation $Ax=b$ has the same solution as the system of equations corresponding to the augmented matrix $\begin{bmatrix}A & b\end{bmatrix}$.

$$\begin{bmatrix}1&2&1&0\\-3&-1&2&1\\0&5&3&-1\end{bmatrix}\xrightarrow[+3\times Row1]{Row2=Row2}\begin{bmatrix}1&2&1&0\\0&5&5&1\\0&5&3&-1\end{bmatrix}\xrightarrow[-Row2]{Row3=Row3}\begin{bmatrix}1&2&1&0\\0&5&5&1\\0&0&-2&-2\end{bmatrix}$$